

CONTACT INFORMATION	329 G, Department of Physics, Indian Institute of Technology Bombay, Mumbai, Maharashtra 400076	ORCID: 0000-0002-5700-282X <i>E-mail:</i> rahulkashyap@iitb.ac.in Website: https://rahulkashyap-phy.github.io/
EDUCATION	• Fall 2012 to Spring 2017 : Distinguished Doctoral Fellow at University of Massachusetts Dartmouth, USA working on my PhD thesis, title “Simulations of Type Ia Supernovae” under the supervision of Prof. Robert Fisher. • Aug 2007–July 2012 : Dual Degree (Masters and Bachelors of Technology) in Ocean Engineering & Naval Architecture at Indian Institute of Technology Kharagpur, India.	
POSITIONS	• Assistant Professor in the Department of Physics, Indian Institute of Technology Bombay, India (since July 2024). • Research Assistant Professor at the Institute of Gravitation and Cosmos and Physics Department at Pennsylvania State University (Sept 2022 - June 2024) • Postdoctoral Fellow at the Institute of Gravitation and Cosmos and Physics Department at Pennsylvania State University (Nov 2019 - Aug 2022). • Max Planck Prize Postdoctoral Fellow at International Centre for Theoretical Sciences (ICTS), Bengaluru, India (Jul 2017-Oct 2019).	
RESEARCH INTERESTS	• Gravitational and electromagnetic wave signals from binary neutron star mergers by detailed GRMHD and radiation hydrodynamics simulations. • Hydrodynamical and magnetohydrodynamical modelling of binary white dwarf mergers and resulting transients such as Type Ia Supernovae and accretion-induced collapse events. • Developing gravitational waves data analysis pipeline to constrain properties of supranuclear matter as well as exotic compact objects from current and future gravitational wave detectors.	
PROFESSIONAL SERVICES	• Peer reviewer for Nature Communications, Physical Review Letters, Astrophysical Journal, and Monthly Notices of Royal Astronomical Society (MNRAS) (https://www.webofscience.com/wos/author/record/3626338) • LIGO Member since October 2017, active in the group responsible for published results of testing general relativity using gravitational wave observations by LIGO-Virgo collaborations. • Member and one of the authors of Cosmic Explorer Horizon Study. • Public outreach of LIGO during the first BNS merger press release. • Involved in various review and analysis activities for collaboration papers from LIGO.	
SELECTED TALKS	• Wednesday Colloquium on loudest binary black hole gravitational wave event, (with Shiraz Minwalla), TIFR, Mumbai, 1 Oct 2025 • 10 years of gravitational Wave Observations, IUCAA Sept 2025. • APS April Meeting 2023: Title: “Systematic errors due to quasi-universal relations in binary neutron stars and their correction for unbiased model selection” • APS April Meeting 2022: Title: “What can we learn from prompt collapse events in binary neutron star mergers? A numerical study” • LIGO Matter group: Effectiveness of universal relations in measuring Neutron Star radius and EOS in 3G GW Detectors • Second Chennai Symposium on Gravitation and Cosmology (CSGC 2022), February 2-5, 2022:	

- Fundamental Theory Seminar, IGC, Penn State: Sept 2021: Impact of prompt and delayed collapse of remnants following binary neutron star mergers.
- APS April Meeting, 2021: Prompt Collapse during Binary Neutron Star Mergers
- Invited talk as Fundamental Theory Seminar on the numerical study of prompt and delayed collapse from BNS mergers at Institute of Gravitation and Cosmos (IGC) at Pennsylvania State University Sept, 2021.
- Prompt Collapse and their implications for properties of NS, APS April Meeting, 2021.
- Standardization of kilonovae and their application cosmology in GR22/Amaldi, Valencia, Spain 2019.
- Short talk and poster presentation on kilonovae standardization and implications on cosmology in GWPAW, University of Maryland, College Park, USA, Dec 2018.
- Standardization of kilonovae in PAX meeting, IUCAA, Pune, India, March, 2018
- Tidal deformability tests of binary black hole mimickers at LVC Meeting Mar 2018, Sonoma State University, California, USA and Sept 2018 Maastricht, Netherlands.
- Summer School on Gravitational Wave Astronomy, ICTS, Bengaluru, July 2017
- Supernovae, Hypernovae and Binary Driven Hypernovae - Adriatic Workshop, ICRANet, Pescara, Italy.
- *The Transient Sky*, The Ninth Harvard-Smithsonian Conference on Theoretical Astrophysics, Sackler Meeting, CfA, Harvard, 2016: Poster Presentation
- Joint Fall Meeting of the APS and AAPT New England Sections, Dartmouth College: Talk
- Fifty-One Erg, 2015, University of North Carolina, Raleigh: A short talk and poster presentation.
- Spring 2015 Meeting of the APS New England Section, Boston University.
- *The Unquiet Universe*, INAF - Astronomical Observatory of Rome, Cefalu, June, 2014 :Poster Presentation.

TEACHING AND
MENTORING
EXPERIENCE

- Numerical Analysis (UG Core) (Autumn 2025), IIT Bombay
- Advanced Astrophysics (Spring 2025, Spring 2026) at IIT Bombay
- First year physics lab (Autumn 2024), IIT Bombay
- Research advisor for three pre-graduate students at ICTS, Bengaluru, India – Pinaki Roy, Sumedha Biswas (now at University of Amsterdam), Abinaya Swaruba (now at Max Planck Institute of Astrophysics).
- Reading course for graduate students on Blackhole Astrophysics at ICTS, Bengaluru, India.
- Full instructor for Precalculus Class to STEM major students in Fall 2016 and Finite Mathematics to business majors in Spring 2017, UMASS Dartmouth.
- Led sections as an instructor (from Spring 2013 to Spring 2016) to physics undergraduates for topics including electrodynamics, waves, optics and modern physics, UMASS Dartmouth.

SCIENCE
OUTREACH

- Demonstration on the formation of cloud to high school students at ICTS, Bengaluru, and at Indian Institute of Science, Bengaluru on National Science Day, 2018 based on one of my undergraduate experimental research works.
- Outreach article covering our work on massive binary white dwarf mergers (check [here](#) and [here](#)).
- Translation of LIGO public release in Hindi for the popularization of gravitational wave science to high school students in India.
- Contributor to the science matter in *Cosmic Zoom*, an outreach program to demonstrate the physics active

at different length scales
[11pt, nofootinbib]revtex4-1

PUBLICATION
LIST

*

References

I. Gupta, Y. Bhoge, **Kashyap, Rahul**, and M. Bhattacharya, *Distinguishing kilonovae from binary neutron star*

and neutron star-black hole mergers (2026), 2603.13484, URL <http://arxiv.org/abs/2603.13484>.

- A. Dhani, A. Camilletti, D. Radice, **Kashyap, Rahul**, B. Sathyaprakash, D. Logoteta, and A. Perego, *Distinguishing prompt-collapse binary neutron star mergers from binary black holes: Tidal effects and remnant properties*, Phys. Rev. D. **112**, 124003 (2025), ISSN 2470-0010,2470-0029, URL <http://link.aps.org/pdf/10.1103/2yng-919s>.
- S. Khadkikar, I. Gupta, **Kashyap, Rahul**, K. Chandra, R. Gamba, and B. S. Sathyaprakash, *Precise and accurate neutron star radius measurements with next-generation gravitational wave detectors*, Phys. Rev. D. **112**, 063020 (2025), ISSN 2470-0010,2470-0029, URL <http://link.aps.org/pdf/10.1103/h6rl-mcjm>.
- Kashyap, Rahul**, I. Gupta, A. Dhani, M. Bapna, and B. Sathyaprakash, *Optimizing bayesian model selection for equation of state of cold neutron stars*, Phys. Rev. D (2025), 2502.03831, URL <http://dx.doi.org/10.1103/kybh-44xf>.
- K. Chandra, I. Gupta, R. Gamba, **Kashyap, Rahul**, D. Chattopadhyay, A. Gonzalez, S. Bernuzzi, and B. S. Sathyaprakash, *On the origins, remnant, and multimessenger prospects of the compact binary merger GW230529*, Astrophys. J. **977**, 167 (2024), ISSN 0004-637X,1538-4357, URL <https://iopscience.iop.org/article/10.3847/1538-4357/ad90bd/meta>.
- R. Huxford, **Kashyap, Rahul**, S. Borhanian, A. Dhani, I. Gupta, and B. S. Sathyaprakash, *Accuracy of neutron star radius measurement with the next generation of terrestrial gravitational-wave observatories*, Phys. Rev. D. **109**, 103035 (2024), ISSN 2470-0010,2470-0029, URL <http://link.aps.org/pdf/10.1103/PhysRevD.109.103035>.
- H. Bandyopadhyay, D. Radice, A. Prakash, A. Dhani, D. Logoteta, A. Perego, and **Kashyap, Rahul**, *Do black holes remember what they are made of?*, Class. Quant. Grav. **41**, 145006 (2024), 2312.11620.
- I. Gupta et al., *Characterizing gravitational wave detector networks: from A[‡] to cosmic explorer*, Class. Quant. Grav. **41**, 245001 (2024), 2307.10421.
- A. Perego, D. Logoteta, D. Radice, S. Bernuzzi, **Kashyap, Rahul**, A. Das, S. Padamata, and A. Prakash, *Probing the incompressibility of nuclear matter at ultrahigh density through the prompt collapse of asymmetric neutron star binaries*, Phys. Rev. Lett. **129**, 032701 (2022), URL <https://link.aps.org/doi/10.1103/PhysRevLett.129.032701>.
- N. C. Roy, V. Tiwari, A. Bobrick, D. Kosakowski, R. Fisher, H. B. Perets, **Kashyap, Rahul**, P. Lorén-Aguilar, and E. García-Berro, *3D hydrodynamical simulations of helium-ignited double-degenerate white dwarf mergers*, ApJL **932**, L24 (2022), ISSN 2041-8205, URL <https://iopscience.iop.org/article/10.3847/2041-8213/ac75e7/meta>.
- Kashyap, Rahul**, A. Dhani, and B. Sathyaprakash, *Systematic errors due to quasiuniversal relations in binary neutron stars and their correction for unbiased model selection*, Phys. Rev. D **106**, 123001 (2022), 2209.02757, URL <http://dx.doi.org/10.1103/PhysRevD.106.123001>.
- Z. Wu, G. Ricigliano, **Kashyap, Rahul**, A. Perego, and D. Radice, *Radiation hydrodynamics modelling of kilonovae with SNEC*, Mon. Not. R. Astron. Soc. **512**, 328 (2022), ISSN 0035-8711, URL <https://academic.oup.com/mnras/article-pdf/512/1/328/42897884/stac399.pdf>.
- Kashyap, Rahul**, A. Das, D. Radice, S. Padamata, A. Prakash, D. Logoteta, A. Perego, D. A. Godzieba, S. Bernuzzi, I. Bombaci, et al., *Numerical relativity simulations of prompt collapse mergers: Threshold mass and phenomenological constraints on neutron star properties after GW170817*, Phys. Rev. D **105**, 103022 (2021), 2111.05183, URL <http://dx.doi.org/10.1103/PhysRevD.105.103022>.

- A. Prakash, D. Radice, D. Logoteta, A. Perego, V. Nedora, I. Bombaci, **Kashyap, Rahul**, S. Bernuzzi, and A. Endrizzi, *Signatures of deconfined quark phases in binary neutron star mergers*, Phys. Rev. D **104**, 083029 (2021), URL <https://link.aps.org/doi/10.1103/PhysRevD.104.083029>.
- N. K. Johnson-McDaniel, A. Mukherjee, **Kashyap, Rahul**, P. Ajith, W. Del Pozzo, and S. Vitale, *Constraining black hole mimickers with gravitational wave observations*, Phys. Rev. D **102**, 123010 (2020), **1804.08026**, URL <https://link.aps.org/doi/10.1103/PhysRevD.102.123010>.
- Kashyap, Rahul**, G. Raman, and P. Ajith, *Can Kilonova Light curves be Standardized?*, Astrophys. J. Lett. **886**, L19 (2019), ISSN 2041-8205,2041-8213, **1908.02168**, URL <http://dx.doi.org/10.3847/2041-8213/ab543f>.
- Kashyap, Rahul**, T. Haque, P. Lorén-Aguilar, E. García-Berro, and R. Fisher, *Double-degenerate carbon-oxygen and oxygen-neon white dwarf mergers: A new mechanism for faint and rapid type ia supernovae*, Astrophys. J. **869** (2018), ISSN 0004-637X, URL <http://dx.doi.org/10.3847/1538-4357/aaedb7>.
- Kashyap, Rahul**, R. Fisher, E. García-Berro, G. Aznar-Siguán, S. Ji, and P. Lorén-Aguilar, *One-Armed Spiral Instability in Double-Degenerate Post-Merger Accretion Disks*, Astrophys. J. **840**, 16 (2017), ISSN 0004-637X, **1704.01584**, URL <http://dx.doi.org/10.3847/1538-4357/aa6afb>.
- P. Dave, **Kashyap, Rahul**, R. Fisher, F. Timmes, D. Townsley, and C. Byrohl, *Constraining The Single-Degenerate Channel of Type Ia Supernovae With Stable Iron-Group Elements in SNR 3C 397*, Astrophys. J. **841**, 58 (2017), ISSN 0004-637X, **1705.02339**, URL <http://dx.doi.org/10.3847/1538-4357/aa7134>.
- D. R. van Rossum, **Kashyap, Rahul**, R. Fisher, R. T. Wollaeger, E. Garcia-Berro, G. Aznar-Siguan, S. Ji, and P. Loren-Aguilar, *Light Curves and Spectra from a Thermonuclear Explosion of a White Dwarf Merger*, Astrophys. J. **827**, 128 (2016), ISSN 0004-637X, **1510.04286**, URL <http://dx.doi.org/10.3847/0004-637X/827/2/128>.
- Kashyap, Rahul**, *Simulations of type ia supernovae (ph.d. thesis)*, Ph.D. thesis, University of Massachusetts Dartmouth (2017).
- Kashyap, Rahul**, R. Fisher, E. García-Berro, G. Aznar-Siguán, S. Ji, and P. Lorén-Aguilar, *Spiral instability can drive thermonuclear explosions in binary white dwarf mergers*, Astrophys. J. Lett. **800**, L7 (2015), ISSN 2041-8205,2041-8213, URL <https://iopscience.iop.org/article/10.1088/2041-8205/800/1/L7>.
- The LIGO Scientific Collaboration and the Virgo Collaboration, *Tests of General Relativity with Binary Black Holes from the second LIGO-Virgo Gravitational-Wave Transient Catalog*, arXiv [gr-qc] (2020), **2010.14529**, URL <http://arxiv.org/abs/2010.14529>.
- The LIGO Scientific Collaboration and the Virgo Collaboration, *Tests of general relativity with the binary black hole signals from the LIGO-virgo catalog GWTC-1*, Phys. Rev. D. **100**, 104036 (2019), ISSN 2470-0010,2470-0029, URL <http://link.aps.org/pdf/10.1103/PhysRevD.100.104036>.